EAS 6134: Inverse Methods in EAS

Shaw problem and Tikhonov regularization

9 February 2026

Shaw diffraction problem (Ex 1.6)

A classic pedagogical inverse problem is an experiment in which an angular distribution of illumination passes through a thin slit and produces a diffraction pattern, for which the intensity is observed (Figure 1.6; [141]).

The data, $d(s)$, are measurements of diffracted light intensity as a function of the outgoing angle $-\pi/2 \leq s \leq \pi/2$. Our goal is to find the intensity of the incident light on the slit, $m(\theta)$, as a function of the incoming angle $-\pi/2 \leq \theta \leq \pi/2$.

The forward problem relating d and m can be expressed as the linear mathematical model,

$$d(s) = \int_{-\pi/2}^{\pi/2} (\cos(s) + \cos(\theta))^2 \left(\frac{\sin(\pi(\sin(s) + \sin(\theta)))}{\pi(\sin(s) + \sin(\theta))} \right)^2 m(\theta) d\theta. \tag{1.26}$$

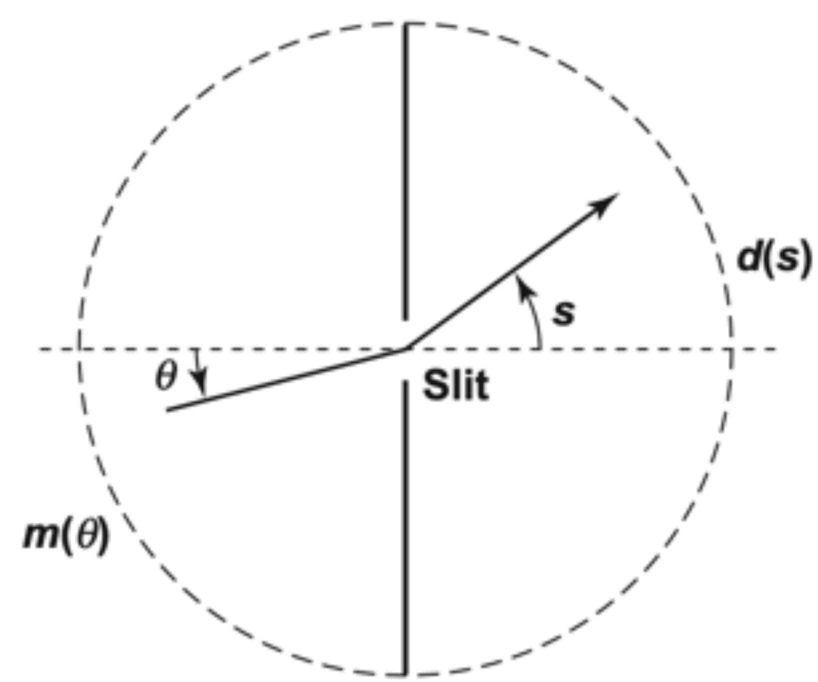

Figure 1.6 The Shaw diffraction intensity problem (1.26).

the data and model points at m and n equally spaced respective angles given by

$$s_i = \frac{(i-0.5)\pi}{m} - \frac{\pi}{2} \quad i = 1, 2, \ldots, m \tag{1.40}$$

and

$$\theta_j = \frac{(j-0.5)\pi}{n} - \frac{\pi}{2} \quad j = 1, 2, \ldots, n. \tag{1.41}$$

Correspondingly, discretizing the data and model into m- and n-length vectors

$$d_i = d(s_i) \quad i = 1, 2, \ldots, m \tag{1.42}$$

and

$$m_j = m(\theta_j) \quad j = 1, 2, \ldots, n \tag{1.43}$$

leads to a discrete linear system $\mathbf{Gm} = \mathbf{d}$, where

$$G_{i,j} = (\cos(s_i) + \cos(\theta_j))^2 \left(\frac{\sin(\pi(\sin(s_i) + \sin(\theta_j)))}{\pi(\sin(s_i) + \sin(\theta_j))} \right)^2 \Delta\theta \tag{1.44}$$

and

$$\Delta\theta = \frac{\pi}{n}. \tag{1.45}$$

Exercise 3.4 (dinosaur problem)

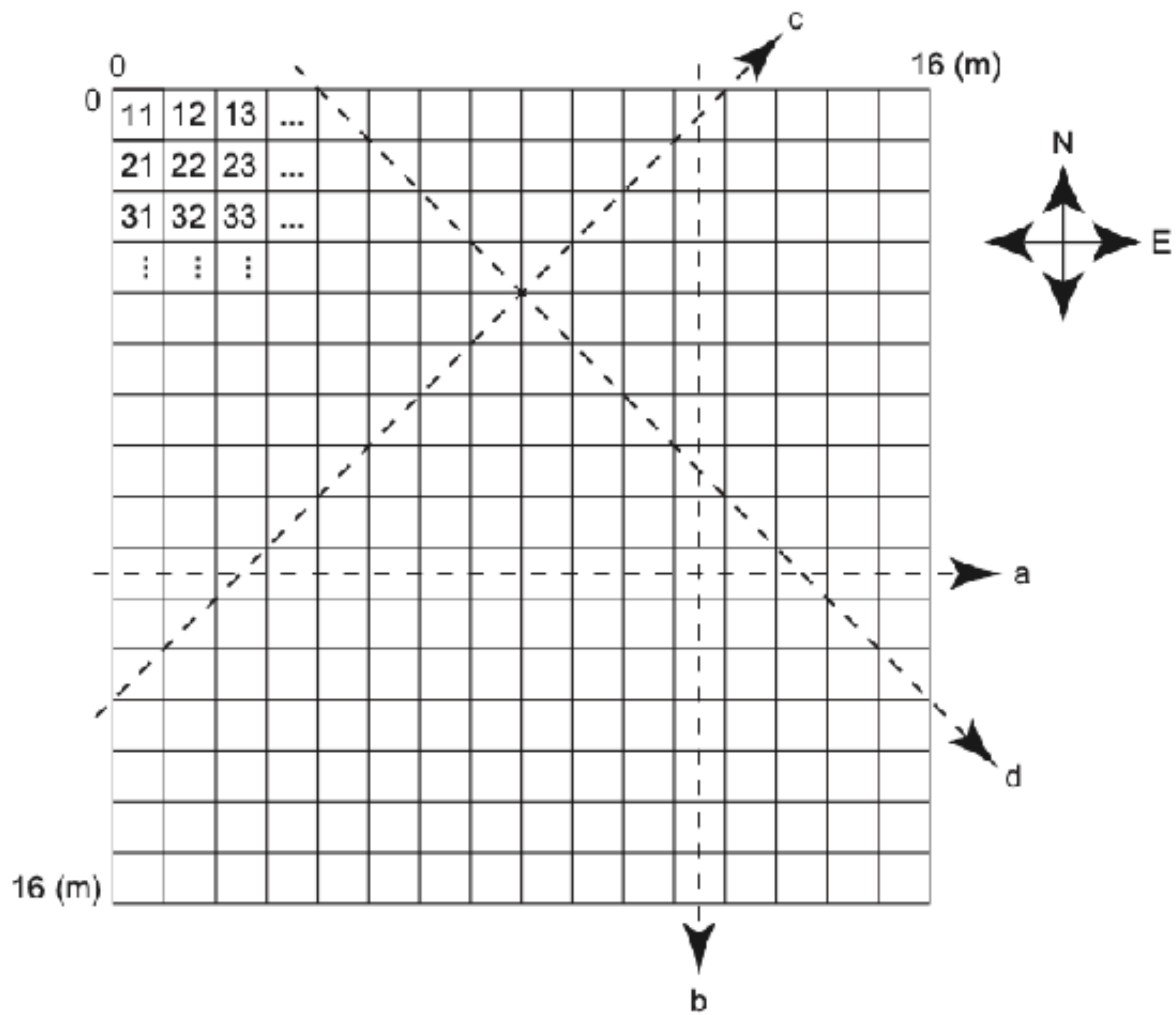

Figure 3.29 Tomography exercise, showing block discretization, block numbering convention, and representative ray paths going east-west (a), north-south (b), southwest-northeast (c), and northwest-southeast (d).

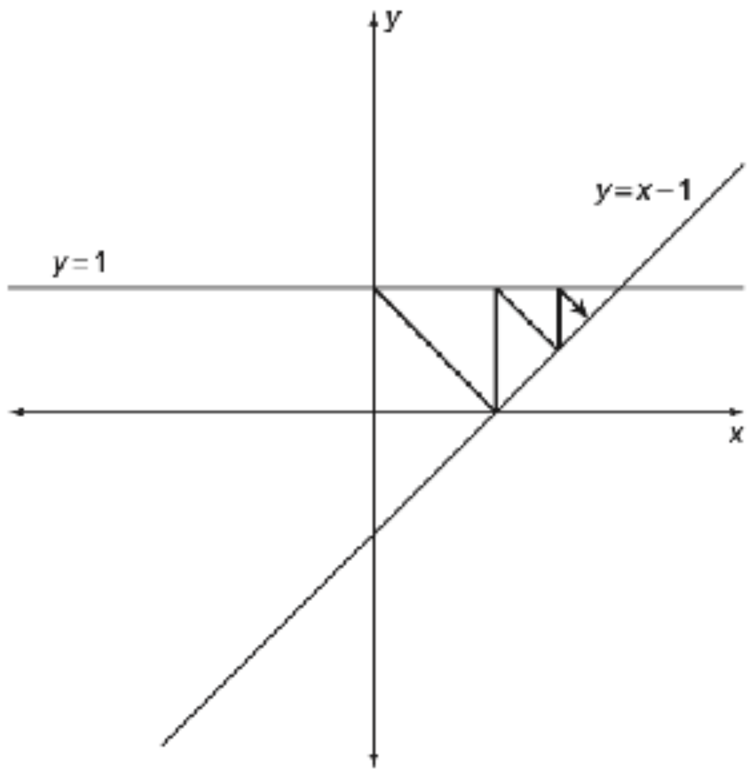

Figure 6.1 Kaczmarz's algorithm on a system of two equations.

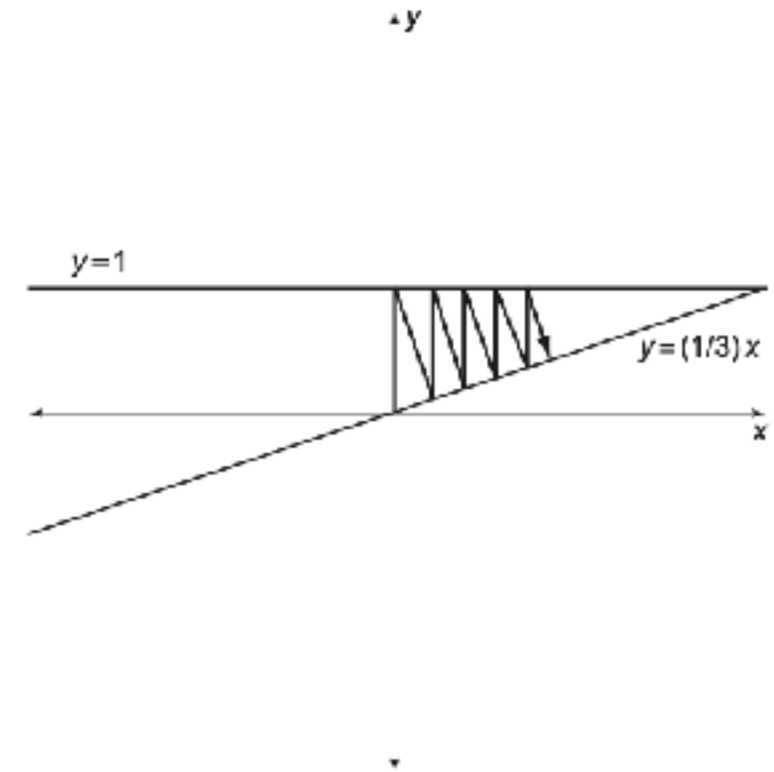

Figure 6.2 Slow convergence occurs when hyperplanes are nearly parallel.

Algorithm 6.1 Kaczmarz's Algorithm

Given a system of equations $\mathbf{Gm} = \mathbf{d}$.

1. Let $\mathbf{m}^{(0)} = \mathbf{0}$.

2. For $i = 0, 1, \ldots, m-1$, let

$$\mathbf{m}^{(i+1)} = \mathbf{m}^{(i)} - \frac{\mathbf{G}_{i+1,\cdot}\mathbf{m}^{(i)} - d_{i+1}}{\mathbf{G}_{i+1,\cdot}\mathbf{G}_{i+1,\cdot}^T}\mathbf{G}_{i+1,\cdot}^T. \tag{6.7}$$

3. If the solution has not yet converged, return to step 2.

Algorithm 6.2 ART

Given a system of equations $\mathbf{Gm} = \mathbf{d}$ arising from a tomography problem:

1. Let $\boldsymbol{m}^{(0)} = \mathbf{0}$.
2. For $i = 0, 1, \ldots, m$, let N_i be the number of cells traversed by ray path i.
3. For $i = 0, 1, \ldots, m$, let L_i be the length of ray path i.

4. For $i = 0, 1, \ldots, m-1, j = 1, 2, \ldots, n$, let

$$m_j^{(i+1)} = \begin{cases} m_j^{(i)} + \frac{d_{i+1}}{L_{i+1}} - \frac{q_{i+1}}{lN_{i+1}} & \text{cell } j \text{ in ray path } i+1 \\ m_j^{(i)} & \text{cell } j \text{ not in ray path } i+1. \end{cases} \tag{6.11}$$

5. If the solution has not yet converged, let $\boldsymbol{m}^{(0)} = \boldsymbol{m}^{(m)}$ and return to step 4. Otherwise, return the solution $\boldsymbol{m} = \boldsymbol{m}^{(m)}$.

The main advantage of ART is that it saves storage. We need only store information about which rays pass through which cells, and we do not need to record the length of each ray in each cell. A second advantage of the method is that it reduces the number of floating-point multiplications compared to that required by Kaczmarz's algorithm. Although in current computers floating-point multiplications and additions require roughly the same amount of time, during the 1970s, when ART was first developed, multiplication was slower than addition.

Algorithm 6.3 SIRT

Given a system of equations $\mathbf{Gm} = \mathbf{d}$ arising from a tomography problem:

1. Let $\mathbf{m}^{(0)} = \mathbf{0}$.
2. For $j = 0, 1, \ldots, n$, let K_j be the number of ray paths that pass through cell j.
3. For $i = 0, 1, \ldots, m$, let L_i be the length of ray path i.
4. For $i = 0, 1, \ldots, m$, let N_i be the number of cells traversed by ray path i.
5. Let $\Delta\boldsymbol{m}^{(i+1)} = \mathbf{0}$.
6. For $i = 0, 1, \ldots, m - 1, j = 1, 2, \ldots, n$, let

$$\Delta m_j^{(i+1)} = \Delta m_j^{(i+1)} + \begin{cases} \frac{d_{i+1}}{L_{i+1}} - \frac{q_{i+1}}{lN_{i+1}} & \text{cell } j \text{ in ray path } i+1 \\ 0 & \text{cell } j \text{ not in ray path } i+1. \end{cases} \tag{6.12}$$

7. For $j = 1, 2, \ldots, n$, let

$$m_j^{(i+1)} = m_j^{(i+1)} + \frac{\Delta m_j^{(i+1)}}{K_j}. \tag{6.13}$$

8. If the solution has not yet converged, return to step 5. Otherwise, return the current solution.

One problem with ART is that the resulting tomographic images tend to be noisier than images produced by Kaczmarz's algorithm (6.7). The simultaneous iterative reconstruction technique (SIRT) is a variation on ART that gives slightly better images in practice, at the expense of a slightly slower algorithm. In the SIRT algorithm, all (up to m nonzero) updates using (6.10) are computed for each cell j of the model, for each ray that passes through cell j. The set of updates for cell j are then averaged before updating the corresponding model element m_j.